

# DAVID GOEDICKE

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I am a research scientist and systems prototyper interested in how to design complex technical systems so people can understand, use, and safely act through them. My work began with building simulation platforms to study interactions with automated systems. Through systems such as StrangeLand, I increasingly treated these platforms as research infrastructure rather than one-off prototypes, examining how to reuse, extend, and support them so others can build on them and conduct more repeatable, comparable research.

This shifted my focus from observing how people use automated systems toward how these systems, and the infrastructure behind them, should be designed to preserve human agency. This includes my work on physical interfaces for meaningful human control, as well as research on reusable software, support structures, and development practices that help researchers both build reliable, technical systems and improve research outcomes.

## EDUCATION

- |                   |                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/2018 – 08/2023 | <b>Cornell Tech</b>   New York City, NY<br>Ph.D. in Information Science<br>Dissertation: <i>DRIVING SIMULATION FOR INTERACTION</i><br>Committee: Wendy Ju, Malte Jung, Serge Belongie |
| 09/2014 – 12/2017 | <b>University of Twente</b>   Enschede, Netherlands<br>M.Sc. in Human Media Interaction                                                                                               |
| 09/2011 – 08/2014 | <b>University of Twente</b>   Enschede, Netherlands<br>B.Sc. Creative Technology<br>Graduated with Cum Laude                                                                          |

## EMPLOYMENT

- |                   |                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|
| 03/2026 – NOW     | <b>UX Interaction Design</b><br>Egosoft<br>Würselen                                                                     |
| 03/2024 – 03/2026 | <b>Post Doctoral Researcher</b><br>Human-Computer Interaction (Prof. Stefan Schneegass)<br>University of Duisburg-Essen |
| 09/2023 – 03/2024 | <b>Post Doctoral Researcher</b>                                                                                         |

|                                |                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
|                                | Future Automation Research Lab (Prof. Wendy Ju)<br>Cornell Tech                                                 |
| 05/2021 – 08/2021              | <b>Mixed Reality Researcher</b> Cornell Tech XR Care Lab with Prof. Deborah Estrin and Harald Haraldson         |
| 05/2020 – 08/2020              | <b>Research and Development Intern</b><br>Toyota Research Institute - Internship<br>ADVISOR: Dr. Hiroshi Yasuda |
| 07/2019 – 08/2019              | <b>Research and Development Intern</b><br>Samsung Electronics · Internship<br>ADVISOR: Prof. Gregory Dudek      |
| 01/2017 – 06/2017              | <b>Visiting Research Scholar</b><br>Stanford University - Center for Design Research<br>ADVISOR: Prof. Wendy Ju |
| 02/2016 – 03/2016              | <b>Visiting Research Scholar</b><br>Nissan Research-SV<br>ADVISOR: Dr. Erik Vinkhuyzen, Melissa Cefkin          |
| 2014 – 2016<br>(project based) | <b>Research Assistant</b><br>Human Media Interaction Group<br>ADVISORS: Prof. Dennis Reidsma, Robby van Delden  |

## INTERNATIONAL COLLABORATIONS

### COLLABORATING LABS

|            |                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| since 2023 | <b>Human Factors &amp; Urban Ergonomics Lab</b> - Research Software<br>Prof. Lynda Ng Boyle<br>NYU, New York, US                                                |
| since 2023 | <b>Carnegie Mellon University Robotics Institute</b> - Research Software<br>Prof. Nikolas Martelaro & Suresh Kumar Jayaraman<br>Pittsburgh, PA, US              |
| since 2023 | <b>Human-Computer Interaction Group &amp; BMW Group</b> Research Software<br>Prof. Andreas Riener, Chantal Himmels<br>Technische Hochschule Ingolstadt, Germany |
| since 2023 | <b>The Georgia Tech Sonification Lab</b> Research Software<br>Prof. Bruce Walker<br>Georgia Tech, Atlanta, US                                                   |
| since 2022 | <b>STUDIO FOR NARRATIVE SPACES</b> Research Collaboration<br>Prof. Ray LC<br>City University Hong Kong                                                          |
| since 2018 | <b>Research Collaboration</b><br>Prof. Avi Parush                                                                                                               |

Technion, Israel  
since 2017 **Center for Design Research** Research Collaboration  
David Sirkin and Becky Currano  
Stanford, California, US

## RESEARCH COLLABORATORS

Prof. Angel Hsing-Chi Hwang (USC, USA), Grace Douglas (NYU, USA), Suresh Kumar Jayaraman (Carnegie Mellon), Prof. Donal Degraen (HIT Lab, New Zealand), Prof. Yuta Sugiura (Keio University, Japan), Prof. (NYU, USA), Prof. Vanessa Evers (University of Twente, Netherlands), Prof. J.D. Zamfirescu-Pereira (UCLA, USA), Hiroshi Yasuda (Toyota Research Institute, USA), Madiha Zahrah Choksi (Cornell Tech, USA), Manaswi Saha (Accenture Labs, USA). Prof. James Grimmelmann (Cornell Tech, USA), Prof. Yan Shvartzshnaider (York University, Canada), Ilan Mandel (Cornell Tech, USA), Prof. Mark Colley (UCL, UK), Dr. Sebastian Feger (LMU, Munich), Prof. Bastian Pfleging (TU Bergakademie, Freiberg), Karoline Brehm (Bauhaus-University, Weimar), Lydia Yanheng Li (City University of Hong Kong), Prof. Valkyrie Savage (University of Copenhagen).

## Technical Expertise

|                                         |                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming                             | Advanced: C, C++, C#, Python, JavaScript, Lua;<br>Basic: Verilog, Rust, Processing                                                                                                                                                                                              |
| Artificial Intelligence                 | Trained smaller models for specific application areas using PyTorch and TensorFlow and deployed local large language models for interaction design experiments and inference pipeline deployment.                                                                               |
| Simulation and VR                       | Using Unity and Unreal, I developed VR/XR simulations and networked multi-user and multi-agent environments for research on autonomous vehicles and robots.                                                                                                                     |
| Automation, Networking<br>and Databases | CICD using GitHub, Linux, Docker, SVN, GitLab.<br>Networking Architectures: REST APIs, Open Sound Control, MQTT.<br>Database: Purpose-built real-time data exchange; visualization and replay of experimental data streams, including integration with large databases via SQL. |
| Physical Computing                      | MCU frameworks (Arduino, PlatformIO, Mbed), Circuit design and manufacturing (EDA), working with FPGAs, sensors, and actuators, as well as CAD, 3D printing, and laser cutting for advanced physical interface design and rapid hardware prototyping.                           |

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|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interaction and Design                | Human-Computer Interaction, interface and interaction design, design thinking, paper prototyping, rapid prototyping.                                                                                                                 |
| Research and Development              | Experimental design, computer simulation, quantitative and qualitative analysis, behavioral research, scenario development, situational-awareness research.                                                                          |
| Collaboration and Research Leadership | Led interdisciplinary academic and industrial research projects, facilitated international collaborations and workshops, and established and maintained research laboratories, including operational leadership during the pandemic. |
| Languages                             | German: Native speaker.<br>English: Full professional proficiency (business fluent).                                                                                                                                                 |

## PUBLICATIONS

**Citations** 513

**h-index** 10

**Conference Papers** 15

**Journal Articles** 1

**Organized workshops** 5

### JOURNAL ARTICLES

**David Goedicke**, Carmel Zolkov, Natalie Friedman, Talia Wise, Avi Parush, Wendy Ju. “Strangers in a Strange Land: New Experimental System for Understanding Driving Culture Using VR”. In: *IEEE Transactions on Vehicular Technology* 71.4 (Apr. 2022), pp. 3399–3413. ISSN: 1939-9359. DOI: [10.1109/TVT.2022.3152611](https://doi.org/10.1109/TVT.2022.3152611).

### CONFERENCE PAPERS (REFEREED)

JiHyun Jeong, **David Goedicke**, Wendy Ju, Guy Hoffman. “Simulating Multiple Road User Perspectives on Autonomous Vehicle Behaviors”. In: *Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '25)*. New York, NY, USA: Association for Computing Machinery, Sept. 2025. ISBN: 979-8-4007-2013-0/2025/09. DOI: [10.1145/3744333.3747817](https://doi.org/10.1145/3744333.3747817). URL: <https://doi.org/10.1145/3744333.3747817>.

Jonathan Liebers, Frieder Sykora, Niklas Pfützenreuter, Uwe Gruenefeld, **David Goedicke**, Stefan Schneegass. “User Identification in Virtual Reality through Behavioral Biometrics and the Influence of Colocated Interactions”. In: *Proceedings of the 2025 31st ACM Symposium on Virtual Reality Software and Technology*. VRST ’25. Association for Computing Machinery, 2025. I S B N: 9798400721182. D O I: [10.1145/3756884.3765978](https://doi.org/10.1145/3756884.3765978). U R L: <https://doi.org/10.1145/3756884.3765978>.

Alia Saad, Jonathan Liebers, Constantin Koczian, Nick Wittig, Roman Heger, Marvin Strauss, Niklas Pfützenreuter, **David Goedicke**, Uwe Gruenefeld, Stefan Schneegass, Donald Degraen. “Investigating Gait Imitation in VR: Impact of Visual Feedback and Avatar Design”. In: *Proceedings of the 24th International Conference on Mobile and Ubiquitous Multimedia*. MUM ’25. Association for Computing Machinery, 2025, pp. 301–309. I S B N: 9798400720154. D O I: [10.1145/3771882.3771912](https://doi.org/10.1145/3771882.3771912). U R L: <https://doi.org/10.1145/3771882.3771912>.

Nick Wittig, Yannick Dohmen, Jonathan Liebers, Donald Degraen, **David Goedicke**, Stefan Schneegass. “FamiliAR Feedback: Investigating Feedback Modality and Familiarity in Classroom Settings Using Spatial Augmented Reality”. In: *Proceedings of the 24th International Conference on Mobile and Ubiquitous Multimedia*. MUM ’25. Association for Computing Machinery, 2025, pp. 273–284. I S B N: 9798400720154. D O I: [10.1145/3771882.3771883](https://doi.org/10.1145/3771882.3771883). U R L: <https://doi.org/10.1145/3771882.3771883>.

Nick Wittig, Noro Schlorke, Roman Heger, Theresa Wettig, Marion Koelle, Uwe Gruenefeld, **David Goedicke**, Donald Degraen, Ricarda Steinmayr, Stefan Schneegass. “You ARe Correct! Comparing Augmented Reality Displays for Individual Feedback in Classroom Settings”. In: *Proceedings of the 24th Interaction Design and Children*. New York, NY, USA: Association for Computing Machinery, 2025, pp. 622–635. I S B N: 9798400714733. U R L: <https://doi.org/10.1145/3713043.3728864>.

Fanjun Bu, Stacey Li, **David Goedicke**, Mark Colley, Gyanendra Sharma, Wendy Ju. “Portobello: Extending Driving Simulation from the Lab to the Road”. In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. CHI ’24. Honourable Mention. Hawai: Association for Computing Machinery, 2024. D O I: [10.1145/3613904.3642341](https://doi.org/10.1145/3613904.3642341). U R L: <https://doi.org/10.1145/3613904.3642341>.

Debargha Dey, **David Goedicke**, Chishang Yang, David Sirkin, Rebecca Currano, Wendy Ju. “GrokWalks: A Portable Virtual Reality Platform to Facilitate Studying Driver-Pedestrian Interactions”. In: *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.

AutomotiveUI '24 Adjunct. Stanford, CA, USA: Association for Computing Machinery, 2024, pp. 284–288. ISBN: 9798400705205. DOI: [10.1145/3641308.3685013](https://doi.org/10.1145/3641308.3685013). URL: <https://doi.org/10.1145/3641308.3685013>.

Rachel DiPirro, Hauke Sandhaus, **David Goedicke**, Dan Calderone, Meeko Oishi, Wendy Ju. “Characterizing Cultural Differences in Naturalistic Driving Interactions”. In: *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*. 2024, pp. 3651–3658. DOI: [10.1109/ITSC58415.2024.10919603](https://doi.org/10.1109/ITSC58415.2024.10919603).

Navit Klein, Hauke Sandhaus, **David Goedicke**, Wendy Ju, Avi Parush. “Modeling Social Situation Awareness in Driving Interactions”. In: *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. AutomotiveUI '24. Stanford, CA, USA: Association for Computing Machinery, 2024, pp. 259–271. ISBN: 9798400705106. DOI: [10.1145/3640792.3675717](https://doi.org/10.1145/3640792.3675717). URL: <https://doi.org/10.1145/3640792.3675717>.

Nick Wittig, Tobias Drey, Theresa Wettig, Jonas Auda, Marion Koelle, **David Goedicke**, Stefan Schneegass. “LeARn at Home: Comparing Augmented Reality and Video Conferencing Remote Tutoring”. In: *Proceedings of the International Conference on Mobile and Ubiquitous Multimedia*. MUM '24. Association for Computing Machinery, 2024, pp. 255–263. ISBN: 9798400712838. DOI: [10.1145/3701571.3701577](https://doi.org/10.1145/3701571.3701577). URL: <https://doi.org/10.1145/3701571.3701577>.

Nialah Jenae Wilson-Small, **David Goedicke**, Kirstin Petersen, Shiri Azenkot. “A Drone Teacher: Designing Physical Human-Drone Interactions for Movement Instruction”. In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. HRI '23. Stockholm, Sweden: Association for Computing Machinery, 2023, pp. 311–320. ISBN: 9781450399647. DOI: [10.1145/3568162.3576985](https://doi.org/10.1145/3568162.3576985). URL: <https://doi.org/10.1145/3568162.3576985>.

**David Goedicke**, Alexandra W.D. Bremers, Sam Lee, Fanjun Bu, Hiroshi Yasuda, Wendy Ju. “XR-OOM: MiXed Reality Driving Simulation with Real Cars for Research and Design”. In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. CHI '22. New Orleans, LA, USA: Association for Computing Machinery, 2022. ISBN: 9781450391573. DOI: [10.1145/3491102.3517704](https://doi.org/10.1145/3491102.3517704). URL: <https://doi.org/10.1145/3491102.3517704>.

Andrea Cuadra, **David Goedicke**, J.D. Zamfirescu-Pereira. “Democratizing Design and Fabrication Using Speech: Exploring Co-Design with a Voice Assistant”. In: *Proceedings of the 3rd Conference on Conversational User Interfaces*. CUI ’21. Bilbao (online), Spain: Association for Computing Machinery, 2021. I S B N: 9781450389983. D O I: [10.1145/3469595.3469624](https://doi.org/10.1145/3469595.3469624). U R L: <https://doi.org/10.1145/3469595.3469624>.

J.D. Zamfirescu-Pereira, David Sirkin, **David Goedicke**, Ray LC, Natalie Friedman, Ilan Mandel, Nikolas Martelaro, Wendy Ju. “Fake It to Make It: Exploratory Prototyping in HRI”. In: *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. HRI ’21 Companion. Boulder, CO, USA: Association for Computing Machinery, 2021, pp. 19–28. I S B N: 9781450382908. D O I: [10.1145/3434074.3446909](https://doi.org/10.1145/3434074.3446909). U R L: <https://doi.org/10.1145/3434074.3446909>.

Jamy Li, Rebecca Currano, David Sirkin, **David Goedicke**, Hamish Tennent, Aaron Levine, Vanessa Evers, Wendy Ju. “On-Road and Online Studies to Investigate Beliefs and Behaviors of Netherlands, US and Mexico Pedestrians Encountering Hidden-Driver Vehicles”. In: *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. HRI ’20. Cambridge, United Kingdom: Association for Computing Machinery, 2020, pp. 141–149. I S B N: 9781450367462. D O I: [10.1145/3319502.3374790](https://doi.org/10.1145/3319502.3374790). U R L: <https://doi.org/10.1145/3319502.3374790>.

Marcel Walch, Stacey Li, Ilan Mandel, **David Goedicke**, Natalie Friedman, Wendy Ju. “Crosswalk Cooperation: A Phone-Integrated Driver-Vehicle Cooperation Approach to Predict the Crossing Intentions of Pedestrians in Automated Driving”. In: *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. AutomotiveUI ’20. Virtual Event, DC, USA: Association for Computing Machinery, 2020, pp. 74–77. I S B N: 9781450380669. D O I: [10.1145/3409251.3411727](https://doi.org/10.1145/3409251.3411727). U R L: <https://doi.org/10.1145/3409251.3411727>.

Vincent Zhang, Natalie Friedman, **David Goedicke**, Dmitriy Rivkin, Michael Jenkin, Xue Liu, Gregory Dudek. “The Answer Is Blowing in the Wind: Directed Air Flow for Socially-acceptable Human-Robot Interaction.” In: *Proceedings International Conference on Robotics, Computer Vision, and Intelligent Systems (ROBOVIS)*. 2020.

Sven Krome, **David Goedicke**, Thomas J. Matarazzo, Zimeng Zhu, Zhenwei Zhang, J. D. Zamfirescu-Pereira, Wendy Ju. “How People Experience Autonomous Intersections: Taking a First-Person Perspective”. In: *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. AutomotiveUI ’19. Utrecht, Netherlands: Association for Computing

Machinery, 2019, pp. 275–283. I S B N: 9781450368841. D O I:  
[10.1145/3342197.3344520](https://doi.org/10.1145/3342197.3344520). U R L:  
<https://doi.org/10.1145/3342197.3344520>.

**David Goedicke**, Jamy Li, Vanessa Evers, Wendy Ju. “VR-OOM: Virtual Reality On-ROad Driving SiMulation”. In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. CHI ’18. Montreal QC, Canada: Association for Computing Machinery, 2018, pp. I–II. I S B N: 9781450356206. D O I:  
[10.1145/3173574.3173739](https://doi.org/10.1145/3173574.3173739). U R L:  
<https://doi.org/10.1145/3173574.3173739>.

#### WORKSHOP PAPERS (REFEREED)

Karoline Brehm, Yan Shvartzshnaider, **David Goedicke**. *Contextual Integrity of A Virtual (Reality) Classroom*. 2023. arXiv: [2303.13684](https://arxiv.org/abs/2303.13684) [cs.CY].

Madiha Zahrah Choksi, **David Goedicke**. *Whose Text Is It Anyway? Exploring BigCode, Intellectual Property, and Ethics*. 2023. arXiv: [2304.02839](https://arxiv.org/abs/2304.02839) [cs.CY].

#### DEMOS, VIDEOS, AND WORK-IN-PROGRESS (REFEREED)

Natalie Friedman, **David Goedicke**, Vincent Zhang, Dmitriy Rivkin, Michael Jenkin, Ziedune Degutyte, Arlene Astell, Xue Liu, Gregory Dudek. “Capturing attention with wind”. In: *Proceedings Workshop on Integrating Multidisciplinary Approaches to Advanced Physical Human-Robot Interaction*. \*Held alongside the IEEE 2020 International Conference on Robotics and Automation (ICRA).

Michael Jenkin, Natalie Friedman, **David Goedicke**, Vincent Zhang, Dmitriy Rivkin, Ziedune Degutyte, Arlene Astell, Xue Liu, Gregory Dudek. “Out of my way! Exploring Different Modalities for Robots to Ask People to Move Out of the Way”. In: *Proceedings Workshop on Active Vision and Perception in Human (Robot) Collaboration*. \*Held alongside the 29th IEEE International Conference on Robot and Human Interactive Communication. September. 2020.

Niklas Pfützenreuter, Carina Liebers, **David Goedicke**, Donald Degraen, Uwe Gruenefeld, Stefan Schneegass. “Eye Want It All! Investigating Eye Tracking as Implicit Support for Generative Inpainting”. In: *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. CHI EA ’26. Association for Computing Machinery, 2026. I S B N: 9798400722813. D O I:  
[10.1145/3772363.3799314](https://doi.org/10.1145/3772363.3799314). U R L:  
<https://doi.org/10.1145/3772363.3799314>.

Nick Wittig, Niklas Pfützenreuter, Carina Liebers, Vinusan Sivalingam, **David Goedicke**, Stefan Schneegass. “GenARate Together: Investigating Collaborative Image Generation Through Tangible Interaction”. In: *Proceedings of the Augmented Humans International Conference 2026*. AHs ’26. Association for Computing Machinery, 2026, pp. 514–525. I S B N: 9798400723513. D O I: [10.1145/3795011.3795013](https://doi.org/10.1145/3795011.3795013). U R L: <https://doi.org/10.1145/3795011.3795013>.

Jonas Keppel, Marvin Strauss, Shuoheng Zhang, Markus Stroehnis, Stefan Lewin, Uwe Gruenefeld, Donald Degraen, **David Goedicke**, Andrii Matviienko, Stefan Schneegass. “The Impact of Bike-Based Controllers and Adaptive Feedback on Immersion and Enjoyment in a Virtual Reality Cycling Exergame”. In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. CHI EA ’25. Association for Computing Machinery, 2025. I S B N: 9798400713958. D O I: [10.1145/3706599.3720096](https://doi.org/10.1145/3706599.3720096). U R L: <https://doi.org/10.1145/3706599.3720096>.

Debargha Dey, **David Goedicke**, Chishang Yang, David Sirkin, Rebecca Currano, Wendy Ju. “GrokWalks: A Portable Virtual Reality Platform to Facilitate Studying Driver-Pedestrian Interactions”. In: *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. AutomotiveUI ’24 Adjunct. Stanford, CA, USA: Association for Computing Machinery, 2024, pp. 284–288. I S B N: 9798400705205. D O I: [10.1145/3641308.3685013](https://doi.org/10.1145/3641308.3685013). U R L: <https://doi.org/10.1145/3641308.3685013>.

**David Goedicke**, Harald Haraldsson, Navit Klein, Lunshi Zhou, Avi Parush, Wendy Ju. “ReRun: Enabling Multi-Perspective Analysis of Driving Interaction in VR”. In: *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. HRI ’23. Stockholm, Sweden: Association for Computing Machinery, 2023, pp. 889–890. I S B N: 9781450399708. D O I: [10.1145/3568294.3580211](https://doi.org/10.1145/3568294.3580211). U R L: <https://doi.org/10.1145/3568294.3580211>.

Regina Bernhaupt, Mark Colley, **David Goedicke**, Alexander Meschtscherjakov, Bastian Pfleging, Andreas Riener, Shadan Sadeghian. “A Critical Perspective on Radically Innovating Personal Mobility”. In: *Adjunct Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. AutomotiveUI ’22. Seoul, Republic of Korea: Association for Computing Machinery, 2022, pp. 215–218. I S B N: 9781450394284. D O I: [10.1145/3544999.3551689](https://doi.org/10.1145/3544999.3551689). U R L: <https://doi.org/10.1145/3544999.3551689>.

## ORGANIZED WORKSHOPS

**David Goedicke**, Stacey Li, Donald Degraen, Kevin John, Yuta Sugiura, Tian Min, Hyunyoung Kim, Sonya S. Kwak, Valkyrie Savage, Tom Igoe, David Sirkin, Wendy Ju, Stefan Schneegass. “Bring Your Own Interface: Exploring Tactile Interaction in Maritime Automation”. In: *Companion Publication of the 2025 ACM Designing Interactive Systems Conference*. New York, NY, USA: Association for Computing Machinery, 2025, pp. 35–38. ISBN: 9798400714863. URL: <https://doi.org/10.1145/3715668.3734183>.

**David Goedicke**, Hamish Tennent, Dylan Moore, Wendy Ju. “Acoustically Aware Robots: Detecting and Evaluating Sounds Robots Make and Hear”. In: *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. HRI ’21 Companion. Boulder, CO, USA: Association for Computing Machinery, 2021, pp. 697–699. ISBN: 9781450382908. DOI: [10.1145/3434074.3444876](https://doi.org/10.1145/3434074.3444876). URL: <https://doi.org/10.1145/3434074.3444876>.

**David Goedicke**, Wendy Ju. “Neural Nets for Music”. In: Stanford University Stanford, CA 94305-8180 USA: Center for Computer Research in Music and Acoustics, 2020. URL: <https://ccrma.stanford.edu/workshops/canceled-using-neural-nets-sonic-interaction-design>.

Nikolas Martelaro, J.D. Zamfirescu-Pereria, **David Goedicke**, David Sirkin, Wendy Ju. “Make This! Introduction to Electronics Prototyping Using Arduino”. In: *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. CHI EA ’20. Honolulu, HI, USA: Association for Computing Machinery, 2020, pp. 1–4. ISBN: 9781450368193. DOI: [10.1145/3334480.3375052](https://doi.org/10.1145/3334480.3375052). URL: <https://doi.org/10.1145/3334480.3375052>.

Sven Krome, Eric Deng, **David Goedicke**, Wendy Ju, Ignacio Alvarez, Jaka Sodnik, Andrew Veit, Francesco Grani. “Simulator Showdown: Pitch Your Virtual Ride”. In: *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications: Adjunct Proceedings*. AutomotiveUI ’19. Utrecht, Netherlands: Association for Computing Machinery, 2019, pp. 25–31. ISBN: 9781450369206. DOI: [10.1145/3349263.3350765](https://doi.org/10.1145/3349263.3350765). URL: <https://doi.org/10.1145/3349263.3350765>.

## PREPRINT ARTICLES

**David Goedicke**, Donald Degraen, Tom Igoe, David Sirkin, Wendy Ju, Stefan Schneegass. *Bring Buttons Back: Physical Interfaces for the Age of Automation*. 2026. arXiv: [2609.14684 \[cs.HC\]](https://arxiv.org/abs/2609.14684). URL: <https://arxiv.org/abs/2609.14684>.

**David Goedicke**, Natalie Chyi, Alexandra Bremers, Stacey Li, James Grimmelmann, Wendy Ju. *Mutual Benefit: The Case for Sharing Autonomous Vehicle Data with the Public*. 2024. arXiv: [2409.01342](https://arxiv.org/abs/2409.01342) [cs.CY]. URL: <https://arxiv.org/abs/2409.01342>.

Karoline Brehm, Yan Shvartzshnaider, **David Goedicke**. *Contextual Integrity of A Virtual (Reality) Classroom*. 2023. arXiv: [2303.13684](https://arxiv.org/abs/2303.13684) [cs.CY]. URL: <https://arxiv.org/abs/2303.13684>.

Madiha Zahrah Choksi, Ilan Mandel, **David Goedicke**, Yan Shvartzshnaider. *Enclosed Loops: How open source communities become datasets*. 2023. arXiv: [2306.05598](https://arxiv.org/abs/2306.05598) [cs.CY].

**David Goedicke**, Mark Colley, Sebastian S. Feger, Michael Goedicke, Bastian Pfleging, Wendy Ju. *Towards Sustainable Research Data Management in Human-Computer Interaction*. 2023. arXiv: [2307.10467](https://arxiv.org/abs/2307.10467) [cs.HC].

## PUBLIC ARTICLES

Manaswi Saha, Wendy Ju, Mike Kuniavsky, **David Goedicke**. *Audio AR: An Introduction*. 2023. URL: <https://medium.com/labs-notebook/audio-ar-an-introduction-698661405ff4>.

## RESEARCH GRANTS & GIFTS

- 2022 National Science Foundation IIS-2212431: **NSF-BSF: HCC: Medium: Cultural Differences in Pedestrian-Autonomous Vehicle Interaction**  
*Investigators:* Wendy Ju (PI), Qian Yang, Cornell (Co-PI), Avi Parush, Technion (Co-PI) – *Assisted in writing grant proposal. Amount:* \$599,986  
*Dates:* 9/01/2022 - 9/30/2024
- 2021 National Science Foundation IIS-2107111: **HCC: Medium: Cultural Differences in Driving Interaction**  
*Investigators:* Wendy Ju (PI), Qian Yang, Cornell (Co-PI), Avi Parush, Technion (Co-PI) – *Assisted in writing grant proposal. Amount:* \$800,000  
*Dates:* 10/01/2021 - 9/30/2024

## INDUSTRIAL PARTNERSHIPS & COLLABORATIONS

- 2020 **Toyota Research Institute** Research and Development of Mixed Reality driving simulators for the development of Autonomous Vehicle features.  
*Investigators:* Wendy Ju & Hiroshi Yasuada

## HONORS & AWARDS

- 2024 **CHI24 “Honorable Mention”** Portobello: Extending Driving Simulation from the Lab to the Road.
- 2017-2018 **Outstanding Teaching Award**, Information Science at Cornell.
- 2012-2014 **Graduated with Cum Laude** BSc. at University of Twente. **B.Sc. Honors program** University of Twente.

## SELECTED PRESS COVERAGE

- 2022 **Cornell Chronicle** *Mixed-reality driving simulator a low-cost alternative*  
<https://news.cornell.edu/stories/2022/04/mixed-reality-driving-simulator-low-cost-alternative>
- 2020 **Cornell Chronicle** *‘Ghostdrivers’ test cultural reactions to autonomous cars*  
<https://news.cornell.edu/stories/2020/04/ghostdrivers-test-cultural-reactions-autonomous-cars>

## TEACHING

- Summer 2025 **Engineering Interactive Systems (Master)**  
Project class focused on prototyping interactive devices with *Prof. Schneegass*.
- Winter 2024 & Summer 2025 **Intelligent User Interfaces (Bachelor)**  
Selected lectures *Prof. Schneegass*.
- Winter 2024 & Summer 2025 **Seminar (Bachelor)**  
Teaching Fundamentals of academic work with *Prof. Schneegass*.
- Spring 2020 **Specialization Project (Masters)**  
Master’s Student Projects at Cornell Tech with *Prof. Estrin* and *Prof. Azenkot*.
- Fall 2019 **Specialization Project (Masters)**  
Master’s Student Projects at Cornell Tech with *Prof. Estrin* and *Prof. Azenkot*.
- Spring 2018 **Developing and Designing Interactive Devices (Masters)**  
I continued to support this class throughout my time at Cornell Tech. *Taught by Wendy Ju*
- 2013–2016 **University of Twente – DesignLab –, Teacher**  
Led substantial parts of the B.Sc. Honours Prototyping and Design course, including student supervision and hands-on teaching. Initiated and hosted Maker Nights, which grew from workshops on augmented reality into an open format for student experimentation.

**Student Supervision** Supervised students from high school to PhD level across diverse backgrounds, from short summer internships to full Master's thesis; selected mentees include: [Nick Wittig](#) [Niklas Pfützenreuter](#), [Carina Liebers](#), [Gavriel Karsch](#), [Raymond Lin](#), [Stav Fahima](#), [Leo Chen](#), [Shayla Lee](#), [Zhaoqi \(Georgia\) Xu](#), [Amithesh Sinha](#), [Justin Harper](#), [Irene Wei](#), [Yasmeen Munasser](#).

## PROFESSIONAL SERVICE

### CONFERENCE ORGANIZING COMMITTEE ROLES

- 2022 **Social Media Chair**  
15th International ACM Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutoUI)
- 2022 **Video Chair**  
ACM Conference on Designing Interactive Systems (DIS)

### SELECTION REVIEWING SERVICE

ACM Conference in Human Factors in Computing (CHI)  
ACM/IEEE Human-Robot Interaction Conference (HRI)  
ACM Conference in Automotive User Interfaces (AutoUI)  
Springer Virtual Reality Journal  
ACM Transactions on Human-Robot Interaction (THRI)  
ACM Transactions on Computer-Human Interaction (TOCHI)  
Mensch und Computer (MuC)  
ACM International Conference on Mobile Human-Computer Interaction (Mobile HCI)  
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)  
International Journal of Human Factors and Ergonomics  
International Journal of Human-Computer Studies  
ACM International Conference Advances in Computer Entertainment Technology (ACE 2016)(now defunct)